Michele Capponi

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Profile

MSc student in Artificial Intelligence and Robotics at Sapienza University of Rome, combining a strong theoretical foundation with practical experience in embedded systems, control engineering, and autonomous robotics. My goal is to contribute to the railway industry, a sector where I see the most compelling intersection of safety, automation, and societal impact. I have applied my skills concretely through embedded software development for UAV platforms, a thesis on Kalman Filtering for battery state estimation, and coursework in real-time systems during an Erasmus exchange at Universidad San Jorge. I bring technical versatility, a collaborative mindset, and a genuine drive to work on complex systems in real operational environments.

Education

Sapienza Università di Roma — Roma, Italy
Master in Artificial Intelligence and Robotics — *Dec. 2024 – Present (expected March 27)*

- **Main topics**: Legged Robots, UAVs, Machine Learning, Deep Learning, Computer Vision, Robotics Manipulators, Autonomous and Mobile Robotics, Robot Programming

Universidad San Jorge — Zaragoza, Spain
Computer Science — *Jan. 2026 – Jun. 2026*

- **Main topics:** Reinforcement Learning, Microservices, Containers, Real Time Operative Systems

Sapienza Università di Roma — Roma, Italy
Bachelor in Statistical Sciences — *Sep. 2021 – Dec. 2024*

- **Main topics**: Probability, Statistical Inference, Statistical Learing, Python and C programming, Calculus, Algebra
- **Thesis:** Kalman Filtering for Battery State of Charge Estimation, project developed for Sapienza Fast Charge mainly through Matlab and Simulink programming

Relevant Experience

Teaching Assistant — April 2026 – Present
Sapienza Università di Roma — *Rome, IT*

- Teaching Assistant for the course of Machine Learning for Signal Processing held in Sapienza
- Python ML Exercises and student support

Robotics Engineer — November 2024 – Present
Sapienza Aerospace Student Association — *Rome, Italy*

- Satellite Embedded Software Development: Python, C++, Git
- Satellite Autonomous Control System: Simulink, C++, Control Engineering
- Satellite Electronic Schematics Designs: Kicad 9.0, Tinkercad

Head of Business and Management Department — Nov. 2023 – Nov. 2025
Sapienza Fast Charge — *Rome, Italy*

- Team Leading: leading and managing a team of 10 students
- Project Management: implementing Agile methodology to support technical projects development
- External Relations: contact and do pitch presentations for possible sponsors

Selected Projects

Kalman Filtering for SoC Estimation (Thesis) | *Matlab, Simulink, Control Theory* — December 2024

- Developed a Simulink Environment for algorithm testing
- Designed and fine-tuned the Kalman Filter algorithms for the battery model

Monocular SLAM | *Matlab (Octave), SLAM, Differential Drive* — June 2026

- Differential Drive localization with monocular camera
- SLAM algorithm development

Humanoid Robot Balance Control | *Python, Optimization, MPC* February 2026

- Humanoid Robot balance control with ZMP
- Model Predictive Control (MPC)

CBF Shared Control for Quadrotors | *Matlab, Shared Control, Simulation* January 2026

- Developed a simulation environment on Matlab for Shared Control of a Quadrotor
- Implemented a Control Barrier Function tested in a life-like environment

Ground to Aerial Image Matching | *Python (PyTorch), Deep Learning, Generative AI* June 2025

- Synthetic Satellite Generation and Joint Feature Learning for Ground-to-Satellite Image Matching
- StableDiffusion Unet for image generation
- Feature Extraction and Learning with VGG and FFN

ML approach Inverse Kinematics of a Robotic Arm | *Python (TensorFlow), classic ML* October 2025

- Machine Learning approach for the control of a robotic arm without knowledge of physical parameters
- Both classical ML techniques and linear Deep Learning approaches

Telegram News bot | *Python, SQLite, RSS, Docker* March 2026

- Developed a telegram bot with automatic RSS feed parsing
- Implemented SQLite database to check for previously sent links
- Developed automation process on Docker and Github Actions

Technical Skills

Languages: Python, C/C++, Matlab/Simulink, HTML
Frameworks and Tools: ROS2, Arduino, Raspberry Pi, Linux, CAD (Solidworks), Gazebo, Docker, Git, Visual Studio Code, Amazon Web Services, FreeRTOS, MQTT, ESP32
Technical skills: Control Theory, Artificial Intelligence (ML, DL, RL), SLAM, Sensor Fusion, AMR, IoT, Front-End, Networks

Certifications and Awards

Teaching Assistant Merit Scholarship

Erasmus Scholarship

Bachelor fully paid by European Union Merit Scholarship

Certification PES PEI PAV **Italian Standard CEI 11-27 / European Standard EN 50110**

BLS-D certification

Kayak Instructor license

Languages

Italian	Native
English	Fluent
Spanish	Fluent
French	Novice

Interests

Trekking and Outdoor Activities: Hiking has developed my resilience and comfort in working in physical environments, qualities I consider essential for field engineering roles

Language Learning: Currently developing French and Portuguese alongside my fluent Italian, English and Spanish, motivated by the international nature of the engineering industry and by cultural interests

DIY Electronics and Making: Personal projects involving microcontrollers and low-level hardware, driven by a genuine curiosity for how systems work at the component level